

Bayesian Probabilistic Selection index

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1 Introduction

This workflow will help you in the Bayesian probabilistic selection index (BPSI) usage. To apply this tool for your crop, it's recommended pay attention to each section and run carefully the codes. Any questions? Contact us by ProbBreed or e-mail.

1.1 Pan-African Soybean Trials dataset

Cultivar recommendation is a critical stage in plant breeding programs, and selecting superior genotypes for multiple traits remains a challenge due to the genotypes $\times$ environment ($G \times E$) interaction.

To address this, we analyzed multi-environment trials data from Pan-African Trials, evaluating 65 soybean genotypes across 19 environments (Araújo et al. 2025).

Our methodological approach integrates Bayesian probabilistic framework to estimate genotypes risk recommendation to each trait and select genotypes according to a multitrait ideotype (Chagas et al. 2025). The Bayesian probabilistic selection index (BPSI) select the 10% genotypes with higher probability of superior performance across environments to Grain Yield (GY), Plant Height (PH) and Number of Days to Maturity (NDM).

1.2 Bayesian probabilistic selection index (BPSI)

The Bayesian probabilistic selection index enables design ideotype by increase or decrease and weighting traits. Along with the probabilities of superior performance across environments (Dias et al. 2022). It accounts for the distance between the worst genotype performance and candidate genotypes for each trait.

$$BPSI_i = \sum_{m=1}^t \frac{\gamma_{pt} - \gamma_{it}}{(1/\lambda_t)}$$

where γ_p is the probability of superior performance of the worst genotype for the trait m , γ is the probability of superior performance of genotype i for trait m , t is the total number of traits evaluated, ($m = 1, 2, \dots, t$), and λ is the weight for each trait t .

2 BPSI example at soybean dataset

In order to provide a step-by-step guide, we applied

2.1 Bayesian Model

The Bayesian model are available at ProbBreed package (Chaves et al. 2024). We used the entry mean model.

$$y_{jk} = \mu + l_k + g_j + \varepsilon_{jk}$$

where the y_{jk} is the phenotypic record of the genotype j^{th} in the k^{th} location, μ is the intercept, l_k is the main effects of the location, g_j is the main effect of the genotype, and ε_{jk} is the residual effect.

```
library(ProbBreed)
```

```
met_df=read.csv("https://raw.githubusercontent.com/tiagobchagas/BPSI/refs/heads/main/Data/blues_long.csv")
```

```
head(met_df)
```

```
##   env gen      PH      GY NDM
## 1 E01 G02 77.56172 3690.918  NA
## 2 E01 G05 81.16229 1004.449  NA
## 3 E01 G09 62.17745 3062.170  NA
## 4 E01 G11 33.37286 1804.674  NA
## 5 E01 G14 52.03038 2719.217  NA
## 6 E01 G21 97.85586 1747.515  NA
```

```
mod = bayes_met(data = met_df,
  gen = "gen",
  loc = "env",
  repl = NULL,
  trait = "PH",
  reg = NULL,
  year = NULL,
  res.het = T,
  iter=400, cores = 4, chain = 4) #recommended run at least 4k iterations
```

```
mod2 = bayes_met(data = met_df,
  gen = "gen",
```

```

        loc = "env",
        repl = NULL,
        trait = "GY",
        reg = NULL,
        year = NULL,
        res.het = T,
        iter = 400, cores = 4, chain = 4) #recommended run at least 4k iterations

mod3 = bayes_met(data = met_df,
                gen = "gen",
                loc = "env",
                repl = NULL,
                trait = "NDM",
                reg = NULL,
                year = NULL,
                res.het = T,
                iter = 400, cores = 4, chain = 4) #recommended run at least 4k iterations

```

2.2 Extracting the probability of superior performance

```

models=list(mod,mod2,mod3)
names(models) <- c("PH","GY","NDM")
inc=c(FALSE,TRUE,FALSE)
names(inc) <- names(models)

results <- lapply(names(models), function(model_name) {
  x <- models[[model_name]] # actual model object

  outs <- extr_outs(model = x,
                    probs = c(0.05, 0.95),
                    verbose = TRUE)

  a <- prob_sup(extr = outs,
                int = .2,
                increase = inc[[model_name]], # + now model_name is a character!
                save.df = FALSE,
                verbose = TRUE)

  return(a)
})
names(results) <- names(models)

```

2.3 Running the BPSI

```

index=bpsi(problist=results,
            increase = c(FALSE,TRUE,FALSE),
            int = 0.1,
            lambda =c(1,2,1),
            save.df = F)

```

2.4 BPSI Ranks

The selected genotypes (G52, G26, G43, G41, G35, G42) have superior performance for the three traits evaluated Grain Yield (GY), Plant Height (PH) and Number of Days to Maturity (NDM).

```
plot(index,category = "Ranks")
```

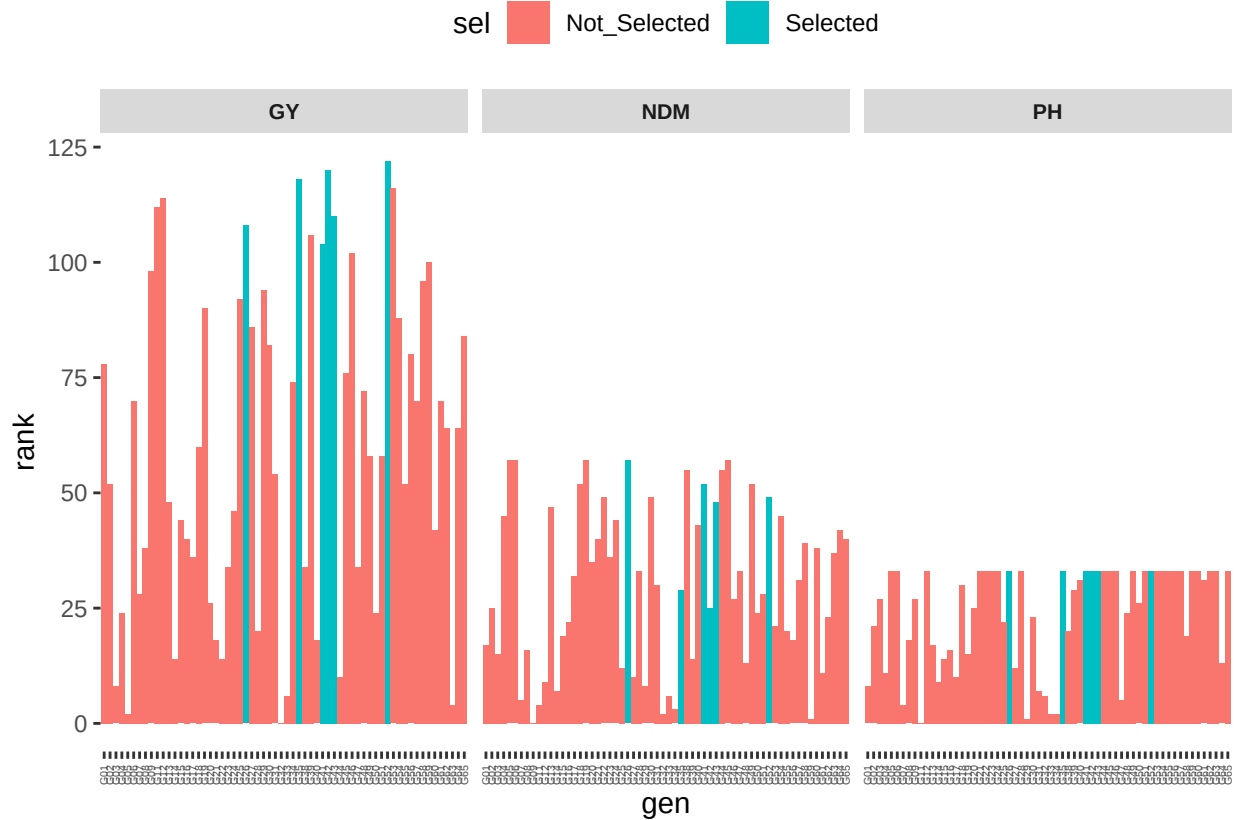


Figure 1: Rank of probability of superior performance of soybean Pan-African Trials across locations in Kenya

2.5 BPSI plot

The selected genotypes (G52, G26, G43, G41, G35, G42) have minor risk of cultivar recommendation to the multitrait ideotype across the Kenya environments.

```
plot(index,category = "BPSI")
```

2.6 Probabilities of superior performance

The probability of superior performance across environments show the risk recommendation of the soybean genotypes being in the top 20% to Grain Yield (GY) and the bottom 20% for Plant Height (PH) and Number of Days to Maturity (NDM).

```
df=index$df
df |> kableExtra::kable()
```

	PH	GY	NDM
G01	0.92250	0.21250	0.25625
G02	0.06375	0.15625	0.10000

	PH	GY	NDM
G03	0.00625	0.11500	0.29500
G04	0.55000	0.13000	0.00750
G05	0.00000	0.10375	0.00000
G06	0.00000	0.17875	0.00000
G07	0.98375	0.13375	0.71125
G08	0.08750	0.13875	0.27625
G09	0.00625	0.25625	0.97625
G11	1.00000	0.35250	0.72500
G12	0.00000	0.35375	0.48250
G13	0.10250	0.15375	0.00625
G14	0.83375	0.11750	0.61625
G15	0.35875	0.15125	0.20875
G16	0.12750	0.14000	0.16500
G17	0.78500	0.13750	0.04500
G18	0.00375	0.17625	0.00250
G19	0.16375	0.23375	0.00000
G20	0.01625	0.13125	0.03375
G21	0.00000	0.12000	0.01500
G22	0.00000	0.11750	0.00375
G23	0.00000	0.13500	0.02750
G24	0.00000	0.15250	0.00875
G25	0.02000	0.23625	0.35625
G26	0.00000	0.33375	0.00000
G27	0.51875	0.23000	0.43125
G28	0.00000	0.12500	0.04125
G29	0.99750	0.24000	0.59125
G30	0.01875	0.21625	0.00375
G31	0.93500	0.15875	0.05875
G32	0.94500	0.10000	0.80875
G33	0.99000	0.10875	0.67250
G34	0.99000	0.18250	0.73750
G35	0.00000	0.40125	0.06125
G38	0.07625	0.13500	0.00125
G39	0.00500	0.30125	0.29625
G40	0.00125	0.12000	0.01000
G41	0.00000	0.28000	0.00250
G42	0.00000	0.47500	0.10000
G43	0.00000	0.34750	0.00500
G44	0.00000	0.11625	0.00125
G45	0.00000	0.18875	0.00000
G46	0.00000	0.27250	0.07625
G47	0.94875	0.13500	0.04125
G48	0.01750	0.18000	0.30125
G49	0.00000	0.16500	0.00250
G50	0.01375	0.13000	0.10500
G51	0.00000	0.16500	0.06625
G52	0.00000	0.52875	0.00375
G53	0.00000	0.39125	0.17375
G54	0.00000	0.23250	0.00750
G55	0.00000	0.15625	0.19500
G56	0.00000	0.21500	0.24500
G57	0.00000	0.17875	0.05250

	PH	GY	NDM
G58	0.08250	0.25250	0.01875
G59	0.00000	0.26125	0.82125
G60	0.00000	0.14500	0.02125
G61	0.00125	0.17875	0.40625
G62	0.00000	0.17750	0.10875
G63	0.00000	0.10750	0.02375
G64	0.42750	0.17750	0.01250
G65	0.00000	0.22000	0.01500

2.7 Genotypes selected

The genotypes by BPSI selected were G52, G26, G43, G41, G35, G42. It have the greater distance to the worst genotype to Grain Yield (GY), Plant Height (PH) and Number of Days to Maturity (NDM). The genotype selected are the 10% multitrait top performance across enviroments.

```
df=index$bpsi

df |> kableExtra::kable()
```

Bibliography

- Araújo, Mauricio S., Saulo Chaves, Gérson N. C. Ferreira, et al. 2025. “High-Resolution Soybean Trial Data Supporting the Expansion of Agriculture in Africa.” *Scientific Data*, ahead of print. <https://doi.org/10.1038/s41597-025-06190-3>.
- Chagas, José T. B., Kaio O. G. Dias, Vinicius Q. Carneiro, et al. 2025. “Bayesian Probabilistic Selection Index in the Selection of Common Bean Families.” *Crop Science* 65 (May): e70072. <https://doi.org/10.1002/CSC2.70072>.
- Chaves, Saulo F. S., Matheus D. Krause, Luiz A. S. Dias, Antonio A. F. Garcia, and Kaio O. G. Dias. 2024. “ProbBreed: A Novel Tool for Calculating the Risk of Cultivar Recommendation in Multienvironment Trials.” *G3 Genes/Genomes/Genetics* 14 (March). <https://doi.org/10.1093/G3JOURNAL/JKAE013>.
- Dias, Kaio O. G., Jhonathan P. R. dos Santos, Matheus D. Krause, et al. 2022. “Leveraging Probability Concepts for Cultivar Recommendation in Multi-Environment Trials.” *Theoretical and Applied Genetics* 135 (April): 1385–99. <https://doi.org/10.1007/S00122-022-04041-Y/FIGURES/4>.

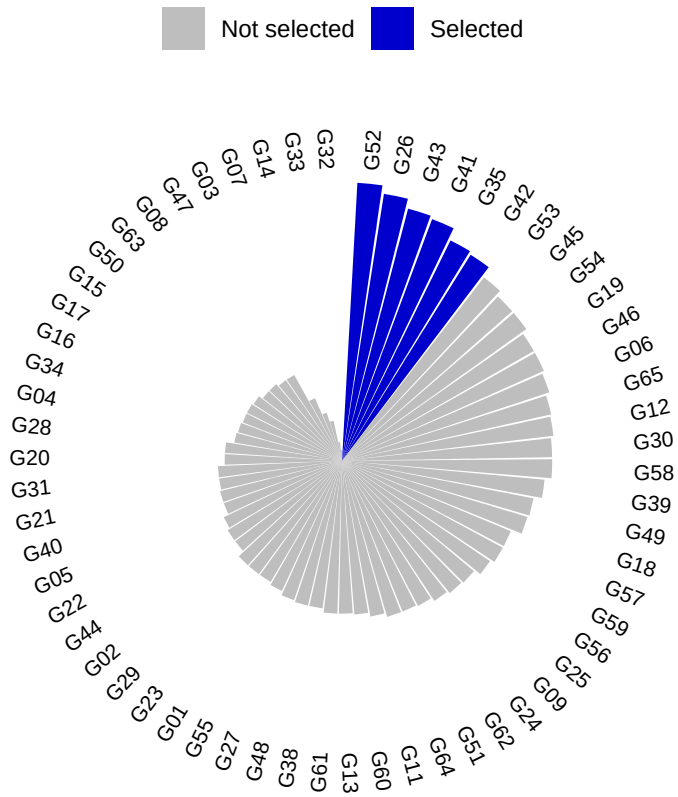


Figure 2: Bayesian probability selection index of soybean genotypes from Pan-African Trials across locations in Kenya